

Curriculum Vitae

Ali Safaya

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Research Interests

Natural Language Understanding, Large Language Models, Entity State Modeling, Memory Augmentation

Education

May 2020—
July 2024 (expected) **Degree:** Doctor of Philosophy in Computer Science
Where: Koç University, Istanbul
GPA: 3.72 of 4.00
PhD research fellow at KUIS AI Center.

August 2015—
June 2019 **Degree:** Bachelor of Science in Computer Engineering
Where: Sakarya University, Sakarya
GPA: 3.89 of 4.00
Valedictorian for the graduation ceremony, for graduating with the highest GPA throughout the degree program.

Research

September 2022—
Present **Project:** Longform Language Models and Neural Memory Augmentation
Where: KUIS AI Center, Koç University, Istanbul
Advisor: Prof. Dr. Deniz Yuret
Contributions: I am working on separating semantic and episodic memory in language models, as well as developing improved memory architectures for modeling long sequences. This is my main research focus for my PhD study.

January 2021—
Present **Project:** Turkish Data Depository
Where: KUIS AI Center, Koç University, Istanbul
Advisor: Prof. Dr. Deniz Yuret
Contributions: I am leading multiple teams of researchers to develop datasets and tools for Turkish NLP. One example of our work was to deliver a new spell-checker to *LibreOffice* team for Turkish language. Also, we built a website to share all available Turkish NLP datasets in an open and documented manner. Another recent project is *Mukayese*: an extensive NLP benchmarking platform for Turkish, which provides test sets and baselines for most of NLP tasks. See tdd.ai.

August 2019—
August 2020 **Project:** ERC funded project of Emerging Markets Welfare
Where: Koç University, Istanbul
Advisor: Dr. Ali Hürriyetoglu
Contributions: I worked as Project Engineer on building a Protest Events Database using Natural Language Processing and Machine Learning, where I gained experience on being a part of a long term research project, and collaborating with interdisciplinary academics. I had the chance to co-organize workshops at top NLP venues like ACL and LREC during my work time on this project.

Employment

April 2021—
Present

Position: ML/NLP Research Scientist
Where: ShallowAI, California
Developing solutions for various ML/NLP applications.

November 2017—
September 2020

Position: ML/NLP Engineer
Where: Casemice Digital, Istanbul
Developing AutoML applications for building multiple persona chat-bots.

Teaching

2020—
Present

Position: Teaching Assistant
Where: Computer Science Dept., Koç University
I worked as a teaching assistant for Deep Learning *COMP-541*, and Natural Language Processing *COMP-542* courses.

Publications

- Ali Safaya and Engin Erzin. “HuBERT-TR: Reviving Turkish Automatic Speech Recognition with Self-supervised Speech Representation Learning”. In: *Computing Research Repository* arXiv:2210.07323v2 (2022). DOI: 10.48550/arXiv.2210.07323. URL: <http://arxiv.org/abs/2210.07323v2>
- Aarohi Srivastava et al. “Beyond the Imitation Game: Quantifying and extrapolating the capabilities of language models”. In: *Computing Research Repository* arXiv:2206.04615 (2022). DOI: 10.48550/ARXIV.2206.04615. URL: <https://arxiv.org/abs/2206.04615>
- Ali Safaya et al. “Mukayese: Turkish NLP Strikes Back”. In: *Findings of the Association for Computational Linguistics: ACL 2022*. Dublin, Ireland: Association for Computational Linguistics, May 2022, pp. 846–863. DOI: 10.18653/v1/2022.findings-acl.69. URL: <https://aclanthology.org/2022.findings-acl.69>
- Ali Hürriyetoğlu et al. “Event Coreference Resolution for Contentious Politics Events”. In: *Computing Research Repository* arXiv:2203.10123 (2022). DOI: 10.48550/ARXIV.2203.10123. URL: <http://arxiv.org/abs/2203.10123>
- Ali Safaya, Moutasem Abdullatif, and Deniz Yuret. “KUISAIL at SemEval-2020 Task 12: BERT-CNN for Offensive Speech Identification in Social Media”. In: *Proceedings of the Fourteenth Workshop on Semantic Evaluation*. Barcelona (online): International Committee for Computational Linguistics, 2020, pp. 2054–2059. URL: <https://aclanthology.org/2020.semeval-1.271>
- Ali Hürriyetoğlu et al. “Automated Extraction of Socio-political Events from News (AESPEN): Workshop and Shared Task Report”. English. In: *Proceedings of the Workshop on Automated Extraction of Socio-political Events from News 2020*. Marseille, France: European Language Resources Association (ELRA), 2020, pp. 1–6. ISBN: 979-10-95546-50-4. URL: <https://aclanthology.org/2020.aespen-1.1>
- Ali Hürriyetoğlu et al. “COVCOR20 at WNUT-2020 Task 2: An Attempt to Combine Deep Learning and Expert rules”. In: *Proceedings of the Sixth Workshop on Noisy User-generated Text (W-NUT 2020)*. Online: Association for Computational Linguistics, 2020, pp. 495–498. DOI: 10.18653/v1/2020.wnut-1.75. URL: <https://aclanthology.org/2020.wnut-1.75>
- Ali Safaya. “Event Sentence Detection Task using Attention model”. In: *Working Notes of CLEF 2019 - Conference and Labs of the Evaluation Forum, 07*. Lugano, Switzerland: CEUR Workshop Proceedings, 2019. URL: http://ceur-ws.org/Vol-2380/paper_122.pdf

Presentations

- Ali Safaya and Taner Sezer. *Turkish Data Depository Project: Towards a Unified Turkish NLP Research Platform*. Seminar presented at KUIS AI meeting, Istanbul (online), 2022
- Ali Safaya. *Reproducing Single Headed Attention - Recurrent Neural Networks*. Seminar presented at KUIS AI meeting, Istanbul (online), 2020

Languages

- Arabic — Native-Bilingual Proficiency
- English — Working Proficiency
- Turkish — Native-Bilingual Proficiency

Technical experience

PyTorch, Jax, CUDA

Honors and awards

- KUIS AI Center Fellowship (2020 — present)
- Turkey's National NLP Hackathon — 1st place award (2021)
- Inzva Computer Engineering PhD Scholarship (2020 — 2021)
- Graduated with Honor Degree of BS program (Valedictorian of 2019)
- Türkiye Scholarship (2015 — 2019)